QP CODE: 2490067

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMIANTION 2024

Multi-Disciplinary-Elective –MG1MDEBCA100-CYBERLAWS AND SECURITY

(2024 ADMISSION ONWARDS)

SCEHEME FOR VALUATION

Maximum Marks: 50

Part-A

Very Short Answer Questions. Answer All Questions

1. Define Cybercrime. [K] [1]

Definition of cybercrime with 2 examples - 2 marks

Cybercrime is criminal activity that use or targets a computer, computer network, or networked device.

Examples: Email and internet fraud, Identity fraud, Theft of financial or card payment data, Cyberextoration, phishing etc.

2. Compare Intellectual property theft and identity theft. [U] [1]

Intellectual property (IP) theft is the unauthorized use, exploitation, or outright theft of creative works, ideas, trade secrets, and proprietary information otherwise protected under intellectual property laws.

Identity theft occurs when someone uses another's personal identifying information, like their name, identifying number, or credit card number, without their permission, to commit fraud or other crimes

3. Explain the role of social media platforms in the prevention of cyberbullying? [U] [1]

Social media platforms play an important role in preventing cyberbullying by implementing various strategies and features to create safer online environments.

Specify *any 2 roles* like *Content Moderation, Educational tools, Anti-Bullying Policies, Community Guidelines, Use of artificial intelligence (AI) and machine learning algorithms employed to detect patterns of bullying behaviour* etc.

4. What do you mean by breach of confidentiality? Which Section deals with it? [K] [1]

A breach of confidentiality is the unauthorized use, disclosure, or access of confidential information and covered by Section 72 of the Information Technology Act, 2000 (IT Act) and the Indian Penal Code (IPC)

5. Explain why password length is often more important than complexity. [An] [2]

Longer passwords make them harder to crack than shorter passwords.

6. What is a hardware token, and how is it used in MFA? [U] [2]

A hardware token is a device that generates one-time codes for authentication purposes.

7. Describe the term "Internet Security." [U] [2]

A central aspect of cyber security, and it includes managing cyber threats and risks associated with the Internet, web browsers, web apps, websites and networks.

8. What role does authentication play in internet security? [K] [3]

Authentication verifies a user's identity, typically through usernames, passwords, or multi-factor authentication.

9. What is cryptology and why is it important in data security? [U] [3]

Cryptology, which includes cryptography and cryptanalysis, is a cyber-security practice that uses complex codes to protect information from unauthorized access.

10. Define Cloud forensics [U] [3]

Cloud forensics is a branch of digital forensics that applies investigative techniques to collecting and evaluating critical evidence in cloud computing environments following a security incident.

(10 X 2 = 20)

Part-B

Short Answer Questions. Answer Any 6 Questions. Each question carries 5 marks

11. Discuss the scope of Cyber laws in regulation of social media and content platforms. [U] [1]

Cyber laws regulate social media by addressing

- Content moderation,
- Privacy, and platform accountability.
- They combat issues like hate speech, cyberbullying, and data misuse
- Protecting intellectual property
- Ensuring transparency in e-commerce Explanation of any three

12. Explain the relevance of the Personal Data Protection Bill and its key feature. [U] [1]

The Personal Data Protection Bill plays a vital role in protecting individuals' privacy by overseeing the collection, storage, and processing of personal data.

Key features include *data localization, user consent requirements, penalties for data breaches, and the establishment of a Data Protection Authority. It empowers users to control their data and ensures accountability for organizations*.

13. What is Multi-Factor Authentication, and why is it important? [U] [2]

Multi-Factor Authentication (MFA) is a security method that requires users to verify their identity using multiple forms, such as password, OTP or security token, or biometric data. It is important because it adds an extra layer of protection, reducing the risk of unauthorized access.MFA is widely used to enhance security in online banking, email, and other sensitive s

14. Explain the benefits of using biometric factors in MFA. [An] [2]

Biometric factors, like fingerprints, are unique to each person and difficult to duplicate Major benefits are

- Enhanced Security.
- Convenience.
- Fraud Prevention

15. List two actions you can take avoid phishing attacks [A] [2]

- Verify Email Sources:

- Use Multi-Factor Authentication (MFA)
- Avoid Sharing Sensitive Information
- Update Software Regularly:
- Enable Spam Filters
- Verify Websites

Explanation of any two

16. Explain the concept of cyber forensic in detail [K] [3]

Cyber forensics, also known as computer forensics, involves the collection, preservation, analysis, and presentation of digital evidence to investigate cybercrimes. It plays a vital role in identifying, analysing, and preserving digital data that can be used as evidence in legal proceedings.

17. How does cryptography ensure the confidentiality, integrity and authenticity of data? [U] [3]

Cryptography ensures confidentiality by ***encrypting data***, making it unreadable to unauthorized users. It maintains integrity through ***hashing,*** where any changes in data can be detected. Authenticity is verified using ***digital signatures***, ensuring the data is from a trusted source. These techniques protect data from unauthorized access, tampering, and impersonation.

18. Illustrate some important system based cyber attacks [U] [3]

- Ransomware
- Phishing
- Malware
- Denial of service attacks (DDoS)
- Password attacks
- SQL injection
- IoT attacks

Explanation of any three. (6 X 5 = 30)

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